



AI Context-Aware Disaster Response Mobile Application

A smart mobile application designed to help people in Myanmar during natural disasters by using Mobile Computing, Ubiquitous Computing, GPS, Edge Computing, Cloud Computing, Artificial Intelligence (AI), and High Performance Computing (HPC).

Project Overview

Natural disasters such as **earthquakes, floods, fires, landslides, and severe weather** can happen unexpectedly and put people's lives at risk. During these emergencies, people often struggle to find trusted information, safe evacuation routes, or quick ways to ask for help.

SafeMyanmar is a smart mobile application that provides emergency assistance before, during, and after disasters. The application focuses on **Myanmar** and aims to provide **trusted disaster information, safe navigation, emergency assistance**, and **AI-powered guidance** to help people stay safe.

The system combines **Mobile Computing, Ubiquitous Computing, GPS, Edge Computing, Cloud Computing, Artificial Intelligence (AI)**, and **High Performance Computing (HPC)** to deliver fast, reliable, and context-aware emergency services.

Project Objectives

The objectives of SafeMyanmar are:

- Provide trusted disaster alerts from verified sources.
 - Help users find the safest evacuation route.
 - Allow users to quickly request emergency assistance.
 - Help rescue teams locate victims using GPS.
 - Provide emergency first aid guidance through AI.
 - Reduce disaster response time.
 - Demonstrate real Mobile Computing and Ubiquitous Computing concepts.
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Target Users

- Citizens
- Families
- Students
- Rescue Teams

- Volunteers
 - Government Disaster Management Organizations
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Main Features

1. Trusted Disaster Alerts

The application receives disaster alerts from **trusted organizations** such as government agencies and disaster management organizations.

Supported disasters include:

- Earthquakes
- Floods
- Fires
- Landslides
- Heavy Rain
- Cyclones
- Severe Weather

Instead of relying on social media rumors, users receive verified disaster information from trusted sources.

Example

 Earthquake detected near Mandalay.

Magnitude: 5.8

Please move to an open area and avoid damaged buildings.

2. GPS Safe Route Navigation

When a disaster occurs, the application automatically detects the user's current GPS location.

It then recommends the safest evacuation route by considering:

- User's current location
- Nearby evacuation shelters
- Blocked roads
- Flooded areas
- Fire danger zones

The application continuously updates the route if conditions change.

3. SOS Emergency Button

If the user is in danger, they can press the SOS button.

The application immediately sends:

- User's GPS location
- Emergency message
- Date and time

to:

- Family members
- Emergency contacts
- Rescue teams (future enhancement)

This allows others to know the user's last known location as quickly as possible.

4. Rescue Beacon Mode

If a user is trapped or cannot move, they can activate **Rescue Beacon Mode**.

The application will:

- Play a loud emergency alarm.
- Flash the phone flashlight in an SOS pattern.
- Display a large **HELP** message on the screen.
- Continue sharing the user's GPS location whenever a network connection is available.
- Enable battery-saving mode to keep the phone operating longer.

This feature makes it easier for nearby people and rescue teams to locate victims.

5. AI Emergency Assistant

The AI Emergency Assistant provides emergency guidance and first aid information.

Users can ask questions such as:

- "What should I do during an earthquake?"
- "How do I stop bleeding?"
- "How should I treat burns?"
- "What should I do if I am trapped?"

If an Internet connection is available:

- The application connects to a cloud AI service for more detailed assistance.

If Internet access is limited:

- The application uses a built-in emergency knowledge base containing first aid instructions and disaster safety guidelines.
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6. First Aid Guide

The application includes emergency first aid information that users can access quickly.

Topics include:

- CPR
- Bleeding control
- Burns
- Fractures
- Earthquake safety
- Flood safety
- Fire safety

Basic first aid information remains available even when Internet access is limited.

7. Trusted Disaster Reports

The application displays disaster reports published by trusted organizations.

Examples include:

- Government disaster management agencies
- Meteorology and weather agencies
- Fire services
- Humanitarian organizations (such as the Myanmar Red Cross Society)

This helps reduce misinformation during emergency situations.

8. Damage Reporting (*Future Feature*)

Users can report disaster damage by submitting:

- Photos
- GPS location
- Description
- Date and time

Examples of reports include:

- Flooded roads
- Fires

- Blocked roads
- Damaged buildings
- Fallen trees

Reports can be reviewed by trusted organizations before being shared with the public.



Mobile Computing Concepts

SafeMyanmar demonstrates several important Mobile Computing concepts.

GPS

- Detects the user's real-time location.
- Finds nearby evacuation shelters.
- Supports safe route navigation.

Mobile Sensors

The application uses smartphone hardware, including:

- GPS
- Camera
- Flashlight
- Accelerometer (future feature)
- Internet connectivity

Wireless Communication

Supports communication using:

- Mobile Network (4G/5G)
- Wi-Fi
- Bluetooth (future enhancement)

Push Notifications

Users receive emergency alerts immediately when new disaster information becomes available.

Mobility

The application continues to support users while they are moving during an emergency.



Ubiquitous Computing Concepts

SafeMyanmar is a **context-aware application**.

The system continuously understands:

- The user's location.
- The current disaster situation.
- The nearest safe shelter.
- The assistance the user may need.

Based on this information, the application automatically provides appropriate recommendations and emergency support without requiring the user to search manually.

Edge Computing

Edge Computing allows some emergency processing to happen closer to the user instead of sending every request to a distant cloud server.

For example:

- Calculate safe evacuation routes.
- Process nearby disaster information.
- Reduce response time during emergencies.

Benefits

- Faster response
 - Lower network delay
 - Reduced cloud traffic
 - Better performance during disasters
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Cloud Computing

Cloud services are responsible for:

- User authentication
- Disaster information database
- AI services
- Real-time data synchronization
- Push notifications
- User report management

The cloud also stores historical disaster data for future analysis.



Artificial Intelligence (AI)

Artificial Intelligence is used to provide:

- Emergency assistance
- Disaster safety recommendations
- Safe route suggestions
- Emergency question answering

Future AI improvements may include:

- Disaster image recognition
- Damage severity prediction
- Personalized emergency recommendations



High Performance Computing (HPC)

High Performance Computing is mainly used on the cloud side to process large amounts of disaster-related information.

Possible applications include:

- Earthquake simulations
- Flood prediction models
- Large-scale evacuation route optimization
- AI model training
- Big data analysis

HPC enables disaster management organizations to process information much faster than traditional computing systems.



System Architecture



